KERALA POLYTECHNIC — REVISION 2026

Course Code 2011 | Semester I | Theory (4 Credits) | Civil Engineering

Building Materials & Construction Techniques

Fundamental building materials, properties, manufacturing, and construction practices for substructures and superstructures.

PROGRAMME	**Civil Engg., Civil & Planning, Rural, Environmental, Construction Technology**
CONTACT HOURS	**60 (L:4 T:0 P:0 R:0)**
CREDITS	**4**
ASSESSMENT	**CIE 40 | ESE 60**
SELF-LEARNING	**96 Hours (Suggested)**

📄 Official Source Declaration

This handbook is based on the official **Kerala Polytechnic Revision 2026** syllabus for **Course 2011 — Building Materials & Construction Techniques**, as provided in the official PDF.

Verified Inconsistencies / Notes:

- The **Self-Learning Activities** table lists 16 weeks × 6 hours = 96 hours, which exceeds the 60 contact hours but is presented as suggested student activities.
- The **ESE Question Paper Pattern** table on Page 5 contains a complex row/column layout. The marks breakdown is: Part A (8×1 = 8), Part B (8×3 = 24), Part C (4×7 = 28), totaling 60. The handbook reproduces this faithfully.
- Series test pattern lists CA5 and CA6 both as "Series Exam (2 modules)" — interpreted as two separate series exams covering two modules each.

📊 Course Dashboard

60
Contact Hours

4
Credits

40+60
CIE + ESE

96
Self-Learning Hours

Teaching & Assessment Scheme

Teaching Scheme

L	T	P	R	Total	Credits	CIE	ESE
4	0	0	0	60	4	40	60

Pre-requisites: 2003A — Chemistry for Engineering Practices (I) & 1011 — Fundamentals of Building Construction (I).

🧭 How to Use This Handbook

📖 **Understand**
Read the detailed explanations, definitions, and classifications for each material and method.

📐 **Visualise**
Study the labelled diagrams, flowcharts, and cross-sections for complex construction techniques.

🧮 **Apply**
Use the built-in calculators and worked examples to solve real-world quantity and design problems.

✅ **Revise**
Test yourself with module-wise questions, the model paper, and the quick-revision cards.

🎯 Course Outcomes & Objectives

Official Course Objectives

1. Impart knowledge on classification, properties, manufacturing, testing, and suitability of basic materials (stone, timber, lime, bricks, blocks).
2. Understand types, properties, and applications of finishing materials, wood products, glass, metals, special purpose, and waste materials.
3. Familiarise with construction methods, site layout, earthwork, and deep foundations.
4. Impart knowledge on types, components, fittings, and functional aspects of doors, windows, ventilators, roofs, trusses, and vertical communication.

Course Outcomes (CO) — Bloom's Level

CO Mapping

CO	Outcome	Bloom

CO	Outcome	Bloom
CO1	Explain classification, properties, uses, manufacturing, testing, and suitability of basic materials.	Understand
CO2	Identify appropriate finishing materials, wood products, glass, metals, and special materials.	Understand
CO3	Describe construction practices for prefabricated structures, site layout, earthwork, and foundations.	Understand
CO4	Explain types, components, and functional requirements of doors, windows, roofs, trusses, and vertical communication.	Understand

CO–PO Mapping Matrix

Course Articulation Matrix (Legends: 1-Low, 2-Medium, 3-High; “-” = No correlation)

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	-	-	-	-	-	-	-	-
CO2	3	2	-	-	-	-	-	-	-	-	-
CO3	3	2	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-
Total	**12**	**6**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**

📋 Curriculum Coverage Map

Module-wise coverage with topics and hours

Module	Title	Topics Covered	Hours	CO
I	Basic Building Materials	Classification, Stone, Timber, Lime, Brick, AAC, Fly ash	18	CO1
II	Flooring, Finishing & Special Materials	Flooring, Wood products, Glass, Metals, Insulation, POP, Waste materials, Special materials	15	CO2
III	Building Construction Practices & Foundation	Prefabricated structures, Site layout, Earthwork, Foundation types	12	CO3
IV	Building Components & Roofing	Doors, Windows, Ventilators, Stairs, Lifts, Escalators, Roofs & Trusses	15	CO4

🛤️ Module Roadmap

📦 Basic Materials
Stone, Timber, Brick

🧱 Finishing & Special
Flooring, Glass, WPC

🏗️ Construction & Foundation
Layout, Earthwork, Piles

🚪 Components & Roofing
Doors, Stairs, Trusses

MODULE I

Basic Building Materials

18 Hours | CO1 | Understand / Remember

📌 **Learning Goals**
Classify materials; describe stone, timber, lime, bricks; explain manufacturing and testing.

🔑 **Key Facts**
Standard brick size: 190×90×90 mm. Fly ash and AAC blocks are eco-friendly alternatives.

🛠️ **Engineering Habit**
Always check field tests (soundness, water absorption) before accepting bricks on site.

Classification of Materials

Building materials are broadly classified into:

- **Natural:** Stone, timber, clay, sand.
- **Artificial:** Bricks, cement, steel, glass.
- **Special:** Waterproofing compounds, fire-resistant materials.
- **Finished:** Tiles, plywood, marble.
- **Recycled:** Fly ash bricks, recycled aggregates.

> **Malayalam support:** [illegible] [illegible] ([illegible], [illegible]), [illegible] ([illegible], [illegible]), [illegible] ([illegible]), [illegible] ([illegible]), [illegible] ([illegible] [illegible] [illegible]) [illegible] [illegible].

Stone — Classification, Requirements, Quarrying & Dressing

Classification of Rocks:

- **Igneous:** Formed from cooling magma (e.g., Granite, Basalt).
- **Sedimentary:** Formed by deposition of sediments (e.g., Sandstone, Lime stone).
- **Metamorphic:** Formed by heat/pressure on existing rocks (e.g., Marble, Slate).

Requirements of good building stone: Hard, durable, strong, free from cracks, fire-resistant, low water absorption, and well-graded.
Quarrying: Extracting stone from natural deposits. Methods: Digging, Wedging, Blasting.
Dressing: Shaping stone to required size and finish. Types: Axed, Hammer-dressed, Chisel-dressed, Rubbed.
Laterite stones: Decomposed form of rocks rich in iron and aluminum; widely used in Kerala, soft when freshly cut but hardens on exposure.

> **Malayalam support:** [illegible] [illegible] [illegible] ([illegible]), [illegible] [illegible] ([illegible]), [illegible] [illegible] ([illegible]) [illegible] [illegible] [illegible]. [illegible] [illegible] [illegible] [illegible] [illegible] [illegible].

Timber — Structure, Properties, Seasoning, Defects & Bamboo

Structure of timber: Outer bark → Inner bark → Cambium → Sap wood → Heart wood → Pith.
Properties: Strong, light, easy to work, good insulator, but susceptible to fire, decay, and insects.
Seasoning: Removing moisture from wood to improve strength and prevent decay.
Methods: **Natural seasoning** (air drying), **Artificial seasoning** (kiln drying, boiling, chemical seasoning).
Defects in timber: Knots, shakes, cracks, warping, fungal decay, insect attack (termites, borers).
Bamboo in construction: Lightweight, renewable, high tensile strength; used for scaffolding, roofing, and low-cost housing.

> **Malayalam support:** [illegible]. [illegible].

Lime — Types and Uses

Lime is obtained by burning limestone ($CaCO_3$).

- **Fat lime (White lime):** High calcium, slakes rapidly, used for plastering.
- **Hydraulic lime:** Contains silica/alumina, sets under water, used for masonry and foundations.
- **Poor lime:** Contains impurities, sets slowly.

Uses: Mortar, plaster, whitewashing, soil stabilisation.

Brick — Constituents, Sizes, Classification, Manufacturing

Constituents of brick earth: Silica (50-60%), Alumina (20-30%), Iron oxide (5-10%), Lime (5-10%), Magnesia, Alkalis.
Sizes: Nominal size = 200×100×100 mm; Standard size = 190×90×90 mm (with 10 mm mortar joints).
Classification of burnt clay bricks: First class, Second class, Third class, Fourth class (based on strength, water absorption, and appearance).
Manufacturing process: Preparation of clay → Moulding → Drying → Burning in kiln (Clamp or Hoffman).
Fly ash bricks & AAC blocks: Made from industrial waste (fly ash) or autoclaved aerated concrete; lightweight, eco-friendly, thermal insulating.

Worked Example: Brick Estimation
Given: Wall size = 10m × 3m × 0.2m; Standard brick size (including mortar) = 0.2m × 0.1m × 0.1m; Wastage = 10%.
Required: Number of bricks.
Calculation: Wall volume = 10 × 3 × 0.2 = 6 m^3. Brick volume = 0.2 × 0.1 × 0.1 = 0.002 m^3. Bricks = 6 / 0.002 = 3000. Add 10% wastage = 3000 × 1.1 = 3300 bricks.
Answer: 3300 bricks.

🧮 **Brick Quantity Estimator**

Wall Length (m)

10

Wall Height (m)

3

Wall Thickness (m)

0.2

Wastage (%)

10

Enter values and click Calculate.

💡 **Module I Tip:** Remember that natural materials like stone and timber are the oldest, while engineered materials like AAC blocks and fly ash bricks are modern, sustainable alternatives. Field tests (sound, water absorption, crushing) are mandatory for bricks on site.

Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

MODULE II

Flooring, Finishing & Special Materials

15 Hours | CO2 | Understand / Remember

📌 **Learning Goals**
Select appropriate flooring, wood products, glass, metals, insulation, and special materials.

🔑 **Key Facts**
Glass is made from sand (SiO_2). M-sand (manufactured sand) is an alternative to river sand.

🛠️ **Engineering Habit**
Choose flooring based on foot traffic, moisture, and aesthetics. Use insulating materials to reduce energy loss.

Flooring Materials

Natural: Granite, Marble — hard, durable, aesthetic.
Artificial: Clay tiles, Ceramic tiles, Pavement blocks — cost-effective, easy to maintain.
Suitability: Granite/Marble for lobbies; ceramic for bathrooms/kitchens; pavement blocks for roads/parks.

Wood Products, Glass, and Metals

Wood products: Plywood (cross-layered), Particle board (wood chips), Veneers (thin slices), Laminated board.
Glass: Soda-lime glass (windows), Lead glass (optical), Toughened glass (safety), Borosilicate (heat resistant).
Metals: Ferrous (Iron, Steel — structural), Non-ferrous (Aluminium, Copper, Zinc — roofing, wiring).

Insulating Materials, POP, Industrial & Agro-Waste

Insulating: Waterproofing (bitumen, PVC), Termite proofing (chemicals), Thermal (glass wool), Sound (acoustic panels).
Plaster of Paris (POP): Calcined gypsum ($CaSO_4 \cdot \frac{1}{2}H_2O$). Uses: plastering, decorative boards, mouldings.
Industrial waste: Fly ash (cement replacement), Blast furnace slag (aggregates).
Agro-waste: Rice husk (insulation), Bagasse (boards), Coir fibres (mats, insulation).

Special Processed Construction Materials

- **Ferro Crete:** Ferrocement — thin, reinforced cement mortar, used for shells and boats.
- **Artificial timber:** Wood-plastic composites, recycled plastics.
- **MDF / HDF:** Medium / High Density Fibreboard — used for furniture.
- **PVC foam board:** Lightweight, waterproof — used for signage and temporary structures.
- **WPC (Wood Plastic Composite):** Durable, weather-resistant — used for decking and fencing.
- **ACP (Aluminium Composite Panel):** Cladding material for facades.
- **Artificial sand (M-sand):** Manufactured from crushed stone, used as fine aggregate.

> 💡 **Module II Tip:** Special materials like WPC and ACP are gaining popularity for their durability and low maintenance. Always check the specific gravity and water absorption of flooring tiles before installation.

MODULE III

Building Construction Practices & Foundation

12 Hours CO3 Understand / Remember

📌 **Learning Goals**
Understand prefabricated structures, site layout, earthwork, and foundation types.

🔑 **Key Facts**
Centre line method is used for load-bearing walls; face line method for framed structures.

🛠️ **Engineering Habit**
Always carry out site clearance and levelling before laying out foundations. Choose foundation based on soil bearing capacity.

Prefabricated Structures & Steel Buildings

Prefabricated structures: Components (columns, beams, slabs) cast in factories and assembled on-site. Advantages: faster construction, quality control, less formwork.
Steel buildings: Structural steel sections (I-beams, channels) used for industrial sheds, high-rise frames. Fast erection, high strength-to-weight ratio.

Site Layout & Earthwork

Job Layout: Marking the building location on ground using centre line and face line methods.
Centre Line Method: Used for load-bearing structures. Total length of centre lines is calculated; earthwork is estimated.
Face Line Method: Used for framed structures. Layout is done by marking column centres and footings.
Excavation: Digging soil for foundations. **Timbering and Strutting:** Supporting trench sides with wooden planks and struts to prevent collapse.
Plinth filling: Filling excavated area with soil/sand up to plinth level to provide a level base.

> **Worked Example: Centre Line Length**
> **Given:** A rectangular building of outer dimensions 12m × 8m, wall thickness 0.3m.
> **Required:** Total centre line length.
> **Calculation:** Centre line length = 2 × (L + W) - 4 × (wall thickness/2) ? Actually, for simple rectangle: 2*(12+8) = 40m. Adjustment for corners: subtract 4 × (wall thickness/2) = 4 × 0.15 = 0.6m. So 40 - 0.6 = 39.4m.
> **Answer:** 39.4m.

Types of Foundations

Shallow Foundation: Depth ≤ width. Types: Strip footing, Isolated footing, Combined footing, Raft footing.
Deep Foundation: Depth > width. Types: Pile foundation, Well foundation (Caisson).
Pile foundation: Load transferred through skin friction and end bearing. Types: Driven piles, Bored piles, Timber piles, Concrete piles.
Well foundation: Large diameter cylindrical caissons sunk to hard strata; used for bridge piers and heavy structures.

> **Malayalam support:** [illegible]

> 💡 **Module III Tip:** The choice between shallow and deep foundation depends on soil bearing capacity. Always conduct a soil test before designing the foundation. Timbering is crucial for safety in deep excavations.

MODULE IV

Building Components & Roofing

15 Hours | CO4 | Understand / Remember

📌 Learning Goals

Understand types of doors, windows, ventilators, staircases, lifts, escalators, and roof trusses.

🔑 Key Facts

Stair rise is usually 150–180mm, tread 250–300mm. King post truss is used for spans up to 7m.

🛠️ Engineering Habit

Design stairs with proper rise/tread ratio for comfort and safety. Provide adequate natural light with windows and ventilators.

Doors — Types and Fittings

- **Fully Paneled / Partly Paneled:** Traditional wooden doors with panels.
- **Flush Doors:** Smooth, core filled with particle board / honeycomb.
- **Collapsible Doors / Rolling Shutters:** Made of steel strips, used for security.
- **Revolving Doors:** For high-traffic commercial buildings.
- **Glazed Doors:** Glass panels for light and visibility.

Fittings: Hinges, handles, locks, bolts, and door closers.

Windows, Ventilators, and Special Types

- **Paneled / Partly Paneled / Glazed:** Wood, Steel, Aluminium frames.
- **Sliding Windows:** Space-saving, easy to operate.
- **Louvered Window:** Angled slats for ventilation and privacy.
- **Bay Window:** Projects outward, enhances view and floor space.
- **Corner Window / Gable / Dormer:** Architectural features for light and aesthetics.
- **Skylight & Ventilators:** Top or high-level openings for natural light and air circulation.

Vertical Communication — Stairs, Lifts & Escalators

Stairs: Types — Straight, L-shaped, U-shaped, Spiral, Dog-legged.
Terms: Tread (horizontal part), Riser (vertical part), Flight, Landing, Handrail.
Lifts: Vertical transport for passengers/goods, powered by electric motors and counterweights.
Escalators: Moving stairs for heavy pedestrian flow in malls and transit stations.

> **Worked Example: Stair Design**
> **Given:** Floor-to-floor height = 3.0m. Rise = 150mm, Tread = 300mm.
> **Required:** Number of risers, treads, and total going.
> **Calculation:** No. of risers = 3000/150 = 20. Treads = 20 - 1 = 19. Total going = 19 × 300 = 5700mm.
> **Answer:** 20 risers, 19 treads, total going = 5.7m.

Roofs and Roof Trusses

Flat roof: Horizontal, used with waterproofing. **Pitched roof:** Sloped, sheds water easily.
Terms: Ridge (top), Eaves (bottom edge), Gable (triangular wall), Hip (sloped end).
Roof trusses: Framed structures supporting the roof. Types: **King Post Truss** (span up to 7m), **Queen Post Truss** (span up to 10m).

> **Malayalam support:** [illegible].
> [illegible].

> 💡 **Module IV Tip:** For residential buildings, a dog-legged staircase with rise 150mm and tread 300mm is comfortable. Ensure that the roof truss is designed to withstand local wind loads and rain.

📐 Formula Bank

Brick Quantity

$N = \frac{V_{wall}}{V_{brick+ mortar}} \times (1 + \text{wastage})$

Centre Line Length

$L = 2(l + w) - 4t/2$

t = wall thickness

Stair Design

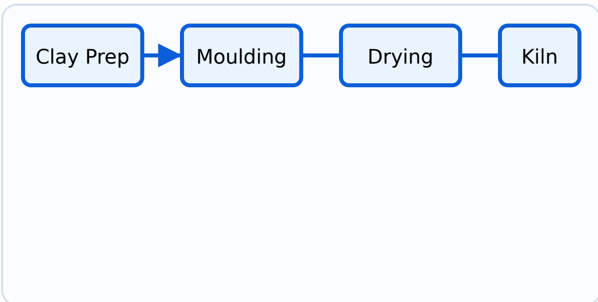
$\text{Risers} = \frac{\text{Height}}{\text{Rise}}$

$\text{Treads} = \text{Risers} - 1$

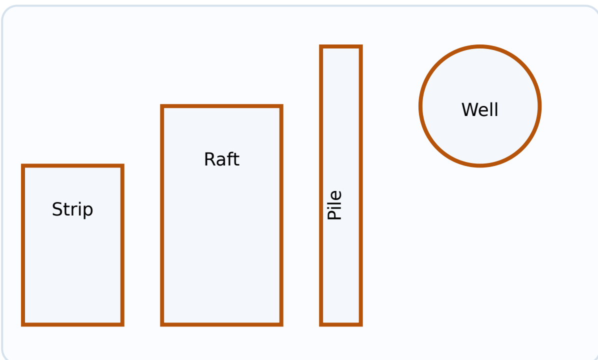
Excavation Volume (Trapezoidal)

$V = \frac{H}{6}(A_1 + A_2 + 4A_m)$

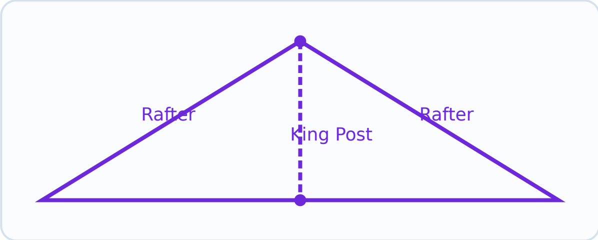
🎨 Visual Library



Brick Manufacturing Process Flow



Foundation Types: Strip, Raft, Pile, Well



King Post Roof Truss

📝 Suggested Self-Learning Activities

Module I

- Study classification of rocks; prepare a report on 'Sustainable materials used in Construction'.
- Case study: Use of bamboo and timber in low-cost or rural construction.
- Compare burnt clay bricks, fly ash bricks, and AAC blocks.
- Collect samples of natural building materials available and prepare a report.

Module II

- Suggest suitable flooring materials for Residential, Hospital, Educational, and Industrial buildings.
- Visit a hardware shop to identify tiles, glass, ACP, PVC boards, plywood, MDF, HDF, etc.
- Measure the quantity of flooring materials used in your home and estimate cost.
- Watch a video on the use of various types of glass used in buildings.

Module III

- Prepare a neat sketch showing centre line and face line method.
- Prepare an estimate of excavation for foundation in your neighbourhood.
- Watch prefabricated building construction videos and summarise advantages.
- Choose suitable foundation type for: Bridge pier, Multi-story building, Black cotton soil.

Module IV

- Study arches — definition, types, materials used for construction.
- Identify fixtures and fastenings for doors and windows.
- Note sizes of doors and windows recommended by BIS.

📊 Assessment Methodology

CIE & ESE Pattern

Mode	Activity	Duration	Marks	Converted
CA1	Self-learning (Module 1)	–	15	5
CA2	Self-learning (Module 2)	–	15	5
CA3	Self-learning (Module 3)	–	15	5
CA4	Self-learning (Module 4)	–	15	5
CA5	Series Exam (2 modules)	2 hours	50	15
CA6	Series Exam (2 modules)	2 hours	50	15
Total CIE				**40**
ESE	End Semester Written Exam	2.5 hours	60	60

ESE Question Paper Pattern
Part A: 2 questions from each module (1 mark each) = 8 marks
Part B: 2 out of 3 questions from each module (3 marks each) = 24 marks
Part C: 1 set question with choice from each module (7 marks each) = 28 marks
Total: 60 marks

❓ Module-wise Questions with Answers

Module I

Q1: Explain the classification of rocks with examples.

Rocks are classified as Igneous (Granite, Basalt), Sedimentary (Sandstone, Limestone), and Metamorphic (Marble, Slate) based on their formation.

Q2: What are the requirements of a good building stone?

It should be hard, durable, strong, free from cracks, fire-resistant, with low water absorption, and well-graded.

Q3: Describe the manufacturing process of burnt clay bricks.

Preparation of clay → Moulding → Drying → Burning in kiln (Clamp or Hoffman kiln).

Module II

Q1: Compare natural and artificial flooring materials.

Natural: Granite, Marble (hard, aesthetic, expensive). Artificial: Ceramic tiles, Pavement blocks (cost-effective, easy to maintain).

Q2: What are the uses of POP and industrial waste materials?

POP: plastering, decorative boards. Fly ash: cement replacement. Slag: aggregates.

Module III

Q1: Explain Centre Line Method and Face Line Method for layout.

Centre line method is used for load-bearing structures by calculating total centre line length. Face line method is for framed structures, marking column centres.

Q2: What are the types of deep foundations?

Pile foundation (driven/bored) and Well foundation (caisson).

Module IV

Q1: List the types of doors and their uses.

Paneled (traditional), Flush (smooth), Collapsible/Rolling shutter (security), Revolving (commercial), Glazed (light).

Q2: Define King Post and Queen Post trusses.

King post: central vertical post, span up to 7m. Queen post: two vertical posts, span up to 10m.

📝 Practice Model Paper

Based on the official pattern — not an official SITTTR model question paper.

Part A (8 × 1 = 8 marks)

1. What is the standard size of a brick?
2. Define seasoning of timber.
3. Name two types of glass.
4. What is M-sand?
5. What is a pile foundation?
6. Define plinth filling.
7. What is a skylight?
8. What is the function of a roof truss?

Part B (8 × 3 = 24 marks)

1. Explain the classification of building materials.
2. Describe the manufacturing process of burnt clay bricks.
3. Compare fly ash bricks and AAC blocks.
4. Explain the uses of industrial and agro-waste materials.
5. Describe the centre line method of layout.
6. Explain types of shallow foundations.
7. Describe the types of stairs.
8. Explain the terms used in pitched roofs.

Part C (4 × 7 = 28 marks)

1. Explain the quarrying and dressing of stone. (7)
2. Explain the properties and uses of timber, along with defects and seasoning methods. (7)
3. Explain site layout, earthwork excavation, and timbering & strutting. (7)
4. Explain types of windows and ventilators with neat sketches. (7)

⚡ Quick Revision

Module I

✔ Materials: Natural/Artificial
✔ Stone: Igneous/Sedimentary/Metamorphic
✔ Brick: 190×90×90 mm, Fly ash/AAC
✔ Timber: Seasoning, Defects

Module II

✔ Flooring: Granite, Marble, Ceramic
✔ Glass: Soda-lime, Toughened
✔ Special: Ferro Crete, WPC, M-sand

Module III

✔ Layout: Centre line / Face line

✔ Foundation: Shallow vs Deep
✔ Pile & Well foundations

Module IV
✔ Doors: Paneled, Flush, Collapsible
✔ Windows: Sliding, Bay, Dormer
✔ Stairs: Rise/Tread, Trusses

Common Mistakes: Confusing nominal and standard brick sizes; mixing up centre line and face line methods; using incorrect rise/tread ratios for stairs; forgetting wastage in brick estimation.

📖 Glossary

Important Terms

Term	Meaning
AAC	Autoclave Aerated Concrete — lightweight, precast blocks.
M-sand	Manufactured sand from crushed stone.
WPC	Wood Plastic Composite — durable, weather-resistant.
POP	Plaster of Paris — calcined gypsum for finishing.
Truss	Framed structure supporting roof.
Riser	Vertical part of a stair.
Tread	Horizontal part of a stair.

📚 References

Textbooks

- Rangawala, S. C. — *Building Construction*, Charotar Publication.
- Ghose, D. N. — *Construction Materials*, Tata McGraw Hill.
- Rangwala, S. C. — *Engineering Materials*, Charotar Publication.

References

- Sood, H. — *Laboratory Manual on Testing of Engineering Materials*, New Age Publishers.

- KBR (Kerala Building Rules) & BIS (Bureau of Indian Standards).

Web Resources (as listed in syllabus)

- (Modules I & II)
- (Module III) (Module IV)

Search terms...